MBERE ML: A Machine Learning Model for Driver-Context Crash-Severity Risk Profiling in the Rwandan Motor Insurance and Fleet Context

BSc. in Software Engineering

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List of Acronyms and Abbreviations

Acronym	Meaning
API	Application Programming Interface
AUC	Area Under the (ROC) Curve
BNR	National Bank of Rwanda (Banque Nationale du Rwanda)
CV	Cross-Validation
ERD	Entity-Relationship Diagram
GDP	Gross Domestic Product
LMIC	Low- and Middle-Income Country
MI	Mutual Information
ML	Machine Learning
RBC	Rwanda Biomedical Centre
RF	Random Forest
RNP	Rwanda National Police
ROC	Receiver Operating Characteristic
RTA	Road Traffic Accident
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
UBI	Usage-Based Insurance
UML	Unified Modeling Language
WHO	World Health Organization
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

Road traffic crashes are a major and growing global public health problem. According to the World Health Organization (WHO, 2023a), approximately 1.19 million people die on the world's roads every year, and road traffic injuries are the leading cause of death for children and young adults aged 5 to 29 years. The burden is highly unequal: about 92% of road fatalities occur in low- and middle-income countries (LMICs), even though these countries hold only around 60% of the world's vehicles, and the WHO African Region records the highest road traffic death rate of any region (WHO, 2023a). These deaths and injuries also carry a heavy economic cost; the World Bank (2018) estimates that LMICs could raise income per capita substantially by as much as 7 to 22% over a 24-year horizon in the countries it studied if they reduced road traffic deaths and injuries, underscoring that road safety is also an economic development issue.

Rwanda reflects this regional pattern. National records show a rising trend in both crashes and deaths: 4,203 crashes with 687 deaths in 2020, 8,639 crashes with 655 deaths in 2021, and 10,334 crashes with 729 deaths in 2022 (Bahati & Masabo, 2025), with the upward trend continuing in 2023 (IGIHE, 2025). Within Rwanda, crashes are heavily concentrated in Kigali, where Patel et al. (2016) mapped distinct road traffic injury hotspots from police data and found that victims were predominantly young males (mean age 35.9 years) and that motorcycles carried an elevated risk of severe injury. Motorcycle taxis are a dominant transport mode in Kigali, which shapes both the nature of crashes and the design of any locally relevant safety solution. an observation echoed in African motorcycle-crash studies elsewhere (Wahab & Jiang, 2019).

Much road-safety analysis treats the crash as the unit of study where crashes happen, how many occur, and how severe they are. A complementary and less studied is which driver, vehicle and environment circumstances are associated with more severe crash outcomes. Insurers and fleet operators, in particular, carry risk at the level of the individual driver, yet in Rwanda they lack data-driven, interpretable tools to characterize that risk. This project, MBERE ML, addresses that gap by developing a machine-learning model that estimates driver-context crash-severity risk from driver, vehicle and environmental features, contextualized for the Rwandan road environment, and delivers the result as decision support through an accessible platform for insurers and fleet operators.

It is important to state at the outset what the available data can and cannot support. Public crash datasets including the Addis Ababa Road Traffic Accident dataset used here (Bedane, 2020), record crashes that occurred; they do not contain a population of drivers with and without crashes, nor exposure measures such as trips, kilometres, policy period or fleet activity. Estimating the prospective probability that a specific active driver will crash requires driver-linked claims or telematics data, which are proprietary and largely unavailable for Rwanda (Pérez-Marín et al., 2019; Pesantez-Narvaez et al., 2019). Accordingly, this project is about crash-severity / driver-context risk profiling among recorded crashes: the model predicts a crash-severity signal (Slight / Serious / Fatal injury) that is used as a driver-context risk proxy, not the future crash probability of an individual driver. This framing is carried consistently through the objectives, research questions, scope and design that follow.

1.2 Problem statement

First, road traffic crashes in Rwanda are frequent, rising and costly. National figures show crashes increasing from 4,203 in 2020 to 10,334 in 2022 (Bahati & Masabo, 2025), and road crashes impose large economic costs on developing economies, removing working-age adults from the labour force and consuming health and productivity resources (World Bank, 2018). Reducing crashes therefore has both a humanitarian and an economic justification, but doing so requires the ability to identify where risk is concentrated.

Second, the institutions that bear the financial consequences of crashes like motor insurers and fleet operators currently have limited data-driven tools for characterising driver-context risk. In practice, motor cover in Rwandan appears to be priced largely through broad risk pooling and coarse categories rather than granular driver-level behavioural risk; motor insurance has been a persistent loss-maker, motorcycle premiums have continued to rise, and in 2026 the sector reported a 21% increase in premiums alongside a national reform effort (The New Times, 2026). We state this cautiously: it is a high-stakes industry claim, and confirming the precise pricing mechanisms would require regulator or National Bank of Rwanda (BNR) reports, insurer pricing manuals, or interviews, which we flag as a data collection requirement rather than an established fact. What is defensible is that a transparent, interpreable driver-context risk signal would provide a useful input for portfolio risk monitoring and for targeting safety interventions.

Third, there is a clear research gap. Machine learning has been applied to Rwandan road safety, but only at an aggregate level: Gatera et al. (2023) forecast short-term accident counts from national police data using Random Forest and Support Vector Machine models, and Bahati & Masabo (2025) used Random Forest and clustering to optimize ambulance placement and predict emergency response time. Neither predicts the risk of an individual driver. More broadly, machine-learning road-safety models for LMICs remain constrained by data scarcity and by the use of variables drawn from high-income settings that omit locally important factors such as unpaved roads, weather and a motorcycle-heavy fleet (Assaye, 2025; Wen et al., 2021).

To address these problems, this project proposes MBERE ML: a machine-learning model that estimates driver-context crash-severity risk from driver, vehicle and environmental features. The target is defined precisely (Section 3.2) as the three-level crash-severity label recorded in the source data: slight, serious and fatal injury used as an ordinal risk proxy. The model is trained and evaluated on real East-African crash data, contextualized for Rwanda's road environment and motorcycle-dominated fleet through feature selection and interpretation, made interpretable through explainable-AI techniques, and delivered through a web platform on which motor insurers and fleet operators can view and analyse severity risk scores as decision support. Because the Addis Ababa dataset is Ethiopian rather than Rwandan, and because crash records differ from insurance/fleet risk data, a domain-shift limitation applies and is treated explicitly throughout this proposal.

1.3 Main Objective

The main objective of this project is to develop and evaluate a machine-learning model that estimates driver-context crash-severity from driver, vehicle and environmental features, contextualized for Rwanda, and to deliver that capability as decision support through an accessible platform for motor insurers and fleet operators. Contextualization is achieved principally through Rwanda-relevant feature engineering and interpretation, and through a synthetic Rwandan-context robustness check; it is not equivalent to validation on real Rwandan data, which is identified as future work should local institutional data become available.

1.3.1 Specific Objectives

1. To build and preprocess a leakage-safe crash-severity dataset. from real East-African police records: the Addis Ababa Road Traffic Accident dataset (Bedane, 2020; 12,316 records, 32 raw columns), clean and engineer at least ten Rwanda-relevant driver/vehicle/environment features (including age, band, driving experience, vehicle type with motorcycle preserved as a first-class category, road surface, light condition, weather and time of the day), A stratified 80/20 hold-out split is reserved and never in training, and feature selection and encoding are fit on the training folds only. The delivery is a document feature contract and a pipeline with a unit-tested no-leakage guarantee.
2. To develop and compare risk-prediction models with rigorous, imbalance-aware evaluation. Compare a transparent rule-based baseline against Random Forest and XGBoost, handling the severe class imbalance (≈84.6% / 14.2% / 1.3% for Slight / Serious / Fatal) with SMOTE applied inside training folds only. Evaluate using macro-F1, macro-recall, per-class recall (especially for the rare Fatal class), precision and one-vs-rest ROC-AUC, with accuracy deprioritized. The measurable success criterion is that the machine-learning models are required to beat the rule-based baseline on the headline metrics (macro-F1, macro-recall, and ROC-AUC.

3. To make predictions interpretable and deliver decision support. Use SHAP for global feature attribution and per-prediction explanations, and deliver predictions through a web platform on which insurers and fleet operators can view, filter and analyse severity-risk scores, per-prediction explanations and a portfolio view, with human-in-the-loop review. The deliverable is a working prototype exposing a risk band, a risk score, class probabilities and a ranked SHAP explanation per prediction.
4. To establish contextual robustness and honest limitations. Generate a synthetic Rwandan-context sample (via Tonic's Fabricate tool) solely to check that the pipeline behaves sensibly on Rwanda-shaped inputs used for functionality and robustness only and never as a source of reported performance metrics and document domain-shift limitations together with a fallback plan should Rwanda National Police (RNP) or Rwanda Biomedical Centre (RBC) data not be obtained before implementation.

1.4 Research Questions

RQ1. Which driver, vehicle and environmental features most strongly associated with crash severity in an East-African LMIC crash dataset relevant to Rwanda?

RQ2. Do machine-learning models (Random Forest, XGBoost) outperform a transparent rule-based baseline for crash-severity risk classification under severe class imbalance, and which model performs best on macro-F1, macro-recall and ROC-AUC?

RQ3. How can severity-risk predictions be made interpretable (via SHAP) and actionable as decision support for Rwandan motor insurers and fleet operators, and what calibration and fairness considerations must accompany their use?

1.5 Project Scope

In scope, the project delivers a data-driven, driver-context crash-severity risk model. It uses the Addis Ababa Road Traffic Accident dataset as the sole training and evaluation source for reported performance, with Rwanda-relevant feature selection; a request to the Rwanda National Police and the Rwanda Biomedical Centre for a local validation sample is made, but the project does not depend on it. The project compares a rule-based baseline against Random Forest and XGBoost models, applies the Synthetic Minority Over-sampling Technique (SMOTE) inside training folds only to address class imbalance, uses SHAP for explainability, and provides a web-platform prototype for the identified beneficiaries: motor insurers and fleet or motorcycle-taxi operators. A synthetic Rwandan-context sample is generated purely as a functionality and robustness check.

The scope is deliberately bounded to protect honesty of claims. The model is first validated on held-out portion of the Addis Ababa crash data; it is not presented as already valid for Rwanda-based driver pricing. The synthetic Rwandan-context data are never reported as performance evidence. Out of scope are on-vehicle sensors or real-time crash detection, nationwide deployment, production integration with insurers' core policy and claims systems, and explicitly any automatic premium-setting function. Because driver-level telematics linked to claims are proprietary and not publicly available, the model characterises risk from the driver, vehicle and environmental attributes recorded in crash data rather than from continuous on-board telematics; collecting primary telematics data is left as future work.

A note on SMOTE and on the beneficiary framing. SMOTE creates synthetic minority samples (Chawla et al., 2002); because synthetic data can distort probability calibration if misapplied (van den Goorbergh et al., 2022), it is used carefully only within training folds, never before the train/split, and alongside class-aware, recall-focused metrics rather than accuracy. Complementary strategies considered include class weighting, threshold tuning and cost-sensitive learning. Regarding beneficiaries, severity-risk scores can influence premiums, employment or access to work, so the prototype is positioned as decision support with human review, disclaimers and fairness checks (Section 3.8), not as a live pricing or operational-decision tool.

1.6 Significance and Justification

The significance of the project is best understood by separating two use cases with different ethical and evidentiary requirements.

Safety intervention and portfolio monitoring (primary). Reducing road crashes carries a strong economic as well as humanitarian case: the World Bank (2018) estimates that substantial gains in income per capita are possible in LMICs that cut road deaths and injuries, so any tool that helps target prevention has measurable value. MBERE ML contributes an interpretable means of flagging higher-severity-risk driver-contexts so that limited safety resources driver feedback, training and check-ins can be focused where they matter most. This use is comparatively easy to justify ethically and academically because it supports human decision-makers rather than replacing them.

Risk-based pricing (secondary). Motor insurance in Rwanda has been under pricing pressure, with rising premiums and a recent reform effort (The New Times, 2026). A driver-context risk signal could, in principle, inform fairer risk-based pricing. However, pricing is regulated, sensitive, and requires actuarial and claims validation that is outside the current data scope; it also raises well-documented fairness and proxy-discrimination concerns (Lindholm et al., 2022; Prince & Schwarcz, 2019). This proposal therefore treats pricing only as a possible downstream application, subject to regulatory approval, actuarial validation and fairness auditing not as a deliverable.

Academically, the project offers a driver-context, behaviour-and-profile-based crash-severity risk model contextualized for Rwanda, moving beyond the aggregate forecasting, hotspot and logistics work that characterizes existing Rwandan research (Gatera et al., 2023; Bahati & Masabo, 2025). It is evaluated on real East-African data rather than imported high-income datasets and is positioned for future Rwandan validation (Assaye, 2025; Wen et al., 2021), with explainability and an explicit fairness and limitations framing as first-class concerns

1.7 Research Budget

Because the project is only software- and data-based, its cost is low. The primary dataset is open and free to access, synthetic-context data are generated with a hosted tool, model training runs on local machines or free cloud tiers, and only modest hosting and connectivity costs are anticipated. All figures below are estimates in United States Dollars (USD) and may vary with the provider and exchange rate.

Table 1: Research budget (estimated project costs).

Item	Qty	Unit cost (USD)	Subtotal (USD)
Open dataset (Addis Ababa RTA, Mendeley Data)	—	0	0
Synthetic-context data generation (Tonic Fabricate free/trial tier)	1	10 (included)	10
Cloud compute (Colab Pro, optional, 2 months)	2	10	20
Platform hosting (prototype, ~6 months)	6	25	150
Internet / data bundles and miscellaneous	—	—	10

1.8 Research Timeline

The Gantt chart in Figure 1 presents an indicative three-month schedule, from the literature review and data acquisition through data preparation, model development, platform build, evaluation and final reporting. The phases overlap deliberately so that modelling and platform work can proceed in parallel once the data are prepared.

Click Here

Figure 1: Research timeline (Gantt chart).

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This review synthesises software and machine-learning-related literature on driver-context accident-risk and crash-severity modelling, crash-frequency forecasting, and the contextualization of such models for low and middle-income countries, with particular attention to work relevant to Rwanda and East Africa. Rather than describing each study in isolation, the review is organised thematically and reads critically across studies: for every theme it identifies what is established, where the methods disagree or fall short, and what the gap implies for MBERE ML.

Literature was identified through keyword searches of indexed platforms Google Scholar, Semantic Scholar, PubMed Central, IEEE Xplore and reputable open-access publishers (for example MDPI, BMC, Springer Nature, PLOS and Elsevier), supplemented by authoritative institutional sources (WHO, World Bank) and the foundational method papers for the techniques used here. Search phrases included "driver accident-risk prediction machine learning," "vehicle crash-severity prediction random forest," "road accident prediction Rwanda," "machine learning road safety low-income country," "class imbalance crash severity SMOTE," "SHAP explainability traffic," and "usage-based insurance driver risk." Sources were screened for relevance, recency and methodological quality; twenty-three sources were retained, of which seventeen are peer-reviewed or scholarly research papers that are analysed below and compared in Table 2.

2.2 Thematic Critical Synthesis

2.2.1 Aggregate Crash forecasting and hotspot mapping

A first tradition models risk at the level of place and time; how many crashes occur, or where hotspots lie typically from historical police records. In Rwanda, Gatera et al. (2023) compared Random Forest and Support Vector Machine regression for short-term forecasting of accident counts, with Random Forest performing best, while Bahati & Masabo (2025) combined Random Forest with clustering and optimization to predict emergency-response time and optimize ambulance placement. Patel et al. (2016) complement these by mapping Kigali injury hotspots from police data, highlighting young male victims and elevated motorcycle risk. Critically, all three are valuable and locally validated, yet they answer a fundamentally different question from the present work: they characterise counts, locations and post-crash logistics, not the driver, vehicle and environmental circumstances associated with a severe outcome. They establish that machine learning is feasible and useful on Rwandan road-safety data, but they leave the driver-context severity question open.

2.2.2 Crash-severity machine learning

A second, larger tradition predicts the severity of a recorded crash from contextual and driver attributes the family this project belongs to. A recent review by Wen et al. (2021) documents the field's shift from statistical models toward tree-based ensembles and notes recurring methodological weaknesses, notably over-reliance on overall accuracy under imbalance and inconsistent minority-class reporting. Concrete studies bear this out. Most directly, Assaye (2025) predicted car-accident severity in Northwest Ethiopia using ten models on 2,000 police-reported accidents integrating driver demographics, behavioural and environmental factors; Random Forest performed best (about 82% accuracy, AUC 0.87), SMOTE was used to address imbalance, and SHAP provided interpretability a close regional and methodological precedent, differing mainly in dataset scale. Shaffiee Haghshenas et al. (2025) compared machine-learning models against a logit baseline and showed that machine learning can classify crash-severity level from real crash records, reinforcing the value of a baseline-versus-ML comparison. Wahab & Jiang (2019) are especially relevant to Rwanda: using Ghanaian national crash data, they modelled motorcycle-crash severity with tree-based algorithms in an African LMIC where two-wheelers dominate directly motivating this project's decision to keep motorcycle as a first-class vehicle category. Read together, these studies converge on tree ensembles (Random Forest, XGBoost) as strong performers on tabular crash data, but they also expose a shared blind spot that this project must confront head-on: headline accuracy can be high while recall on the rarest, most consequential (fatal) class remains poor, so accuracy is a misleading yardstick under severe imbalance.

2.2.3 Driver-level and telematics / usage-based insurance risk

A third tradition, used mainly by insurers, models risk at the level of the individual driver, either from declared attributes or from telematics capturing behaviour such as speed, braking and time of day. Pesantez-Narvaez et al. (2019) compared XGBoost with logistic regression for predicting motor-insurance claims from telematics, finding that the added interpretability of the simpler model can rival gradient boosting for rare-event claim prediction a caution against assuming that a more complex model is automatically better on imbalanced insurance data. Pérez-Marín et al.

(2019) used telematics-based quantile regression to assess the risk of driving above posted speed limits, illustrating the granularity that genuine behavioural data permit. The critical lesson for MBERE ML is a data lesson: true prospective per-driver risk prediction depends on driver-linked behaviour, claims and exposure data, which are proprietary and effectively unavailable for Rwanda, and whose anonymized or foreign feature sets resist local contextual interpretation. This is precisely why the present work is scoped to severity/risk-profiling from interpretable crash records rather than to claim-probability prediction positioning it honestly between the severity-modelling and driver-risk traditions.

2.2.4 LMIC contextualization and data scarcity

A cross-cutting theme is that models built for high-income settings transfer poorly to LMICs. WHO (2023a, 2023b) and the World Bank (2018) establish both the scale of the LMIC burden and its concentration in the African Region. Assaye (2025) and Wen et al. (2021) argue that imported feature sets omit locally decisive factors unpaved roads, weather exposure, and a motorcycle-heavy fleet while Wahab & Jiang (2019) demonstrate the value of models built on African data with locally meaningful variables. The critical implication is doubled. First, contextualization for Rwanda is achievable through feature engineering (preserving motorcycle, road surface, light and weather conditions) and interpretation, even without Rwandan training data. Second, and honestly, using an Ethiopian dataset introduces domain shift relative to Rwanda, and crash records differ from insurance/fleet data; neither can be assumed away, which is why the project reports Addis-only performance and treats Rwandan applicability as a validation requirement rather than a claim.

2.2.5 Class imbalance and explainability: methods

The methodological backbone of the project rests on well-established techniques, but the literature also flags their pitfalls. Random Forest (Breiman, 2001) and XGBoost (Chen & Guestrin, 2016) are the tree-ensemble workhorses that repeatedly top crash-severity benchmarks. For imbalance, SMOTE (Chawla et al., 2002) remains the default oversampler, yet van den Goorbergh et al. (2022) show that imbalance corrections such as SMOTE can harm probability calibration models become over-confident about the minority class. This is a decisive, critical finding for a risk system: it argues for applying SMOTE only inside training folds, for foregrounding ranking and recall metrics over accuracy, and for treating the resulting scores as relative rankings rather than calibrated probabilities unless calibration is explicitly checked. For interpretability, SHAP (Lundberg & Lee, 2017) provides a theoretically grounded, additive attribution framework, and its tree-specific extension (Lundberg et al., 2020) makes exact, efficient explanations feasible for Random Forest and XGBoost the exact combination adopted here for both global feature importance and per-prediction explanations.

2.2.6 Fairness and responsible use in insurance and risk scoring

Finally, a scoring model that touches insurance and employment cannot ignore fairness. Lindholm et al. (2022) analyse discrimination and fairness in insurance pricing and show that merely dropping protected attributes is insufficient, because non-protected features can act as proxies so-called indirect or proxy discrimination. Prince & Schwarcz (2019) develop the same concern from a legal-policy standpoint for AI and big-data systems. For MBERE ML this is directly actionable: driver attributes such as age band and sex can proxy protected characteristics, which motivates (i) dropping weak and sensitive inputs during feature selection, (ii) auditing outcomes across age, sex, vehicle type and the motorcycle category, and (iii) restricting the system to human-reviewed decision support rather than automatic pricing. These commitments are formalised in the ethics design of **Section 3.8**.

2.3 Comparative Analysis Matrix

Table 2 compares the principal reviewed studies along dimensions that matter for this project: their prediction focus and target, their data and context, their method, and critically the gap or lesson each contributes. The final column makes explicit that the datasets closest to this work support crash-severity, not individual crash-probability, targets.

Table 2: Comparative analysis matrix of reviewed literature.

Author / source (year)	Focus, data and context	Method	Key finding and critical lesson for MBERE ML
Gatera et al. (2023)	Accident-count forecasting; RNP national data; Rwanda	Random Forest, SVM regression	RF forecasts counts best; aggregate, not driver-context different question, but proves local ML feasibility.
Bahati & Masabo (2025)	Response time and ambulance siting; Rwanda accident + EMS data	Random Forest + clustering + optimization	Post-crash logistics, not pre-crash risk; confirms RF suitability on Rwandan data.
Patel et al. (2016)	Injury-hotspot epidemiology; Kigali police data	Spatial hotspot analysis	Young-male, motorcycle-heavy risk profile motivates keeping motorcycle first-class.
Assaye (2025)	Crash-severity; 2,000 police records; NW Ethiopia (LMIC)	10 ML models + SMOTE + SHAP	RF best (~82% acc, AUC 0.87); closest regional precedent; but accuracy misleads under imbalance.
Shaffiee Haghshenas et al. (2025)	Crash-severity level; real crash records	ML vs. logit baseline	ML beats logit justifies a transparent baseline the ML models must beat.
Wahab &	Motorcycle-crash	Tree-based ML	African, two-wheeler-focused

Author / source (year)	Focus, data and context	Method	Key finding and critical lesson for MBERE ML
Jiang (2019)	severity; national data; Ghana (LMIC)	(J48, IBk, RF)	severity ML; strongest motivation for motorcycle handling and local features.
Wen et al. (2021)	Review of ML for crash-severity modelling	Systematic review	Field moved to tree ensembles; warns of accuracy over-reliance and weak minority reporting.
Pesantez-Narvaez et al. (2019)	Motor-claim prediction; telematics; insurer data	XGBoost vs. logistic regression	Simpler model can rival boosting for rare claims; complexity is not automatically better.
Pérez-Marín et al. (2019)	Driving-above-limit risk; telematics	Quantile regression	Shows granularity that genuine telematics permit unavailable for Rwanda, hence severity focus.
Breiman (2001); Chen & Guestrin (2016)	Method foundations (tabular ML)	Random Forest; XGBoost	The two ensemble workhorses adopted here; strong on tabular crash data.
Chawla et al. (2002)	Class-imbalance method	SMOTE oversampling	Default minority oversampler; must be applied inside training folds to avoid leakage.
van den Goorbergh et al. (2022)	Imbalance corrections and calibration	Simulation with logistic regression	SMOTE can harm calibration prefer recall/ranking metrics; treat scores as rankings unless calibrated.
Lundberg & Lee (2017);	Explainability	SHAP; TreeExplainer	Grounded additive attributions; exact, efficient for RF/XGBoost

Author / source (year)	Focus, data and context	Method	Key finding and critical lesson for MBERE ML
Lundberg et al. (2020)			adopted for global and local explanations.
Lindholm et al. (2022); Prince & Schwarcz (2019)	Fairness in insurance / AI pricing	Actuarial and legal analysis	Proxy discrimination: dropping protected attributes is not enough audit outcomes; no automatic pricing.
Bedane (2020)	Primary dataset; 12,316 Addis records; Ethiopia	Police-record dataset	Interpretable driver/road/environment features; sole source of reported metrics; target = severity.

2.4 Strengths, Weaknesses and Research Gap

Table 3 distils the strengths and Rwanda-specific weaknesses of each approach, clarifying the niche MBERE ML occupies.

Table 3: Strengths and weaknesses of existing approaches for the Rwandan context.

Approach	Strengths	Weaknesses / gaps for Rwanda
Aggregate crash forecasting (Gatera et al., 2023)	Locally validated; useful for planning and resource allocation.	Predicts counts and hotspots, not which driver-contexts are severe-risk.
Crash-severity ML (Assaye, 2025; Shaffiee Haghshenas et al., 2025; Wahab & Jiang, 2019)	Accurate; interpretable driver/environment features; African LMIC precedent exists.	Predicts severity of recorded crashes, not prospective driver risk; often reports accuracy on imbalanced data with weak fatal-class recall.

Approach	Strengths	Weaknesses / gaps for Rwanda
Telematics / usage-based insurance models (Pérez-Marín et al., 2019; Pesantez-Narvaez et al., 2019)	Genuine driver-level behavioural risk; matches insurer use case.	Require proprietary telematics linked to claims; no public Rwandan data; anonymized features resist local interpretation.
Ambulance / response optimization (Bahati & Masabo, 2025)	Improves emergency-response logistics in Rwanda.	Post-crash focus; does not characterise driver-context risk before a crash.

Research gap. No published Rwandan study characterises driver-context crash-severity risk using locally relevant, interpretable features; Rwandan machine learning to date is aggregate (counts, hotspots, logistics), while the interpretable severity tradition sits mostly in other LMICs and the genuine driver-risk tradition depends on data Rwanda does not have. MBERE ML addresses this gap modestly and honestly: it moves toward driver-context severity-risk modelling using the best available East-African crash data, adds explainability and a fairness framing, and sets out the requirements for true Rwandan driver-level validation as future work.

2.5 General Comment and Conclusion

The reviewed literature confirms that the components of MBERE ML are individually well established tree ensembles for tabular crash data, SMOTE for imbalance (applied with care for calibration), and SHAP for explanation and that the chosen direction is well grounded. It also disciplines the project's claims. Machine learning is already applied to Rwandan road safety, but only at the aggregate level; interpretable crash-severity modelling with imbalance handling has been demonstrated on real East-African data; and the insurance sector's genuine driver-risk framing depends on telematics and claims data that Rwanda lacks publicly. MBERE ML is therefore positioned not as a solution that fully closes the driver-risk gap, but as a step that develops driver-context severity-risk modelling on available East-African crash data, delivers it as interpretable decision support, and specifies explicitly what would be required: Rwandan institutional data, exposure information, and fairness/actuarial validation to extend it toward true Rwandan driver-level risk and pricing.

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter describes the methodology and design of the MBERE ML system. The work follows a Design Science Research orientation building and evaluating an artefact that solves a real problem combined with an iterative, incremental development model so that the data pipeline, machine-learning models and web platform can be built and tested in short, verifiable cycles. The research is primarily experimental: data are prepared and engineered, models are trained and compared against measurable metrics, and the resulting predictions are delivered through a prototype platform for the intended beneficiaries.

3.2 Target Definition and Sampling Strategy

3.2.1 Prediction target

The prediction target is the recorded crash-severity label, Accident_severity, a three-class ordinal variable with classes Slight injury (encoded 0), Serious injury (1) and Fatal injury (2). This is a multiclass classification task used as a driver-context risk proxy: the model estimates how severe a crash outcome tends to be given a driver/vehicle/environment context, and does not claim to predict whether a specific active driver will crash in the future. Defining the target this way resolves the central ambiguity raised in review the label is crash severity, not crash involvement or claim filing and aligns every downstream design choice (metrics, imbalance handling, risk score) with what the data can support.

3.2.2 Data sources and their roles

Primary source (sole source of reported metrics). The Addis Ababa Road Traffic Accident dataset (Bedane, 2020) comprises 12,316 real crash records with 32 raw columns of interpretable driver, vehicle, road and environmental attributes and the severity target. A stratified 80/20 hold-out test set (2,464 records) is the single source of all reported performance figures.

Synthetic Rwandan-context sample (never a metric source). A synthetic sample of Rwanda-shaped driver-context profiles is generated with Tonic Fabricate solely to check that the trained pipeline accepts and behaves sensibly on Rwanda-relevant inputs (functionality and contextual-robustness testing). It is documented in the code, and repeated here, that this synthetic data is never used to compute or report model performance; doing so would be misleading, and the design deliberately forbids it.

3.2.3 Feature inclusion, exclusion and selection

Raw columns are renamed and bucketed into interpretable, Rwanda-relevant features. Feature selection uses mutual information against the target, computed on the training folds only, with a low threshold; the whole-feature granularity preserves interpretability (one-hot groups are kept or dropped together). Two candidate attributes driver sex and driver education carried the weakest signal and were pruned, which has the additional benefit of removing a direct protected-attribute input (sex) from the model, consistent with the fairness commitments in **Section 3.8**. Table 4 below lists the engineered features and their treatment.

Table 4: Engineered features and their treatment in the model.

Feature	Encoding	Role / rationale
driver_age_band	Ordinal	Kept: strongest individual signal; audited for fairness.
driver_experience	Ordinal	Kept: experience is a plausible risk correlate.
time_of_day	Ordinal (Night/Morning/Afternoon/Evening)	Kept: derived from crash hour; interpretable exposure proxy.
vehicle_service_year	Ordinal	Kept: vehicle condition proxy.
vehicle_type	One-hot (Motorcycle first-class)	Kept: motorcycle preserved as its own class for the Rwandan fleet.
weather / road_surface / light_condition	One-hot	Kept: locally decisive environmental factors.
driver_vehicle_relation / vehicle_owner / vehicle_defect	One-hot	Kept: modest but non-trivial signal.
driver_sex	(candidate)	Pruned: weak signal; also removes a direct protected attribute.
driver_education	(candidate)	Pruned: weak signal.

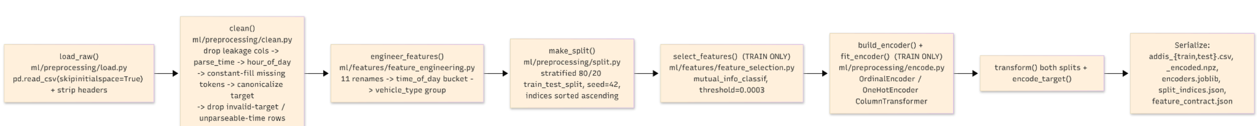
An honest observation, surfaced by the analysis rather than hidden, is that individual features carry weak predictive signal (mutual-information scores are small across all candidates). This is expected: the data are profile-level crash records without telematics or exposure, which caps the achievable individual-level signal. It is reported as a finding, not a defect, and reinforces the decision to prefer recall- and ranking-based evaluation over accuracy.

3.2.4 Class distribution, splitting and fallback plan

The severity classes are severely imbalanced; approximately 84.6% Slight, 14.2% Serious and 1.3% Fatal, an imbalance ratio near 66:1. Validation is two-tier: a stratified 80/20 hold-out that is never touched during training, plus stratified 5-fold cross-validation within the training split for model comparison. SMOTE is applied only inside the cross-validation training folds (and on the full training set for the final refit), never on validation or test data; this leakage-safe ordering is a unit-tested invariant of the pipeline. Should Rwandan institutional data from RNP or RBC not be obtained before implementation, the fallback plan is explicit: the model is trained and evaluated entirely on the Addis Ababa data, and Rwandan validation is treated as future work or as qualitative expert validation, with no change to the honesty of the reported results.

3.3 Research Design and Proposed Model

The proposed methodology is summarized in Figure 2. Real crash-record data from the Addis Ababa dataset are cleaned with deterministic, parameter-free rules (constant-fill for missing tokens; dropping rows with an invalid target or unparseable time), so cleaning is safe to run before the split. Rwanda-relevant features are then engineered, after which the stratified 80/20 split is performed. Only on the training side are feature selection and encoding fit; the single most important guard against learning any statistic from the test set. A transparent rule-based baseline is then compared against Random Forest (Breiman, 2001) and XGBoost (Chen & Guestrin, 2016), both of which are wrapped so that SMOTE (Chawla et al., 2002) resamples only within each training fold. SHAP (Lundberg & Lee, 2017; Lundberg et al., 2020) explains the predictions, the model emits a severity-risk score and band, and the result is served through a web platform.



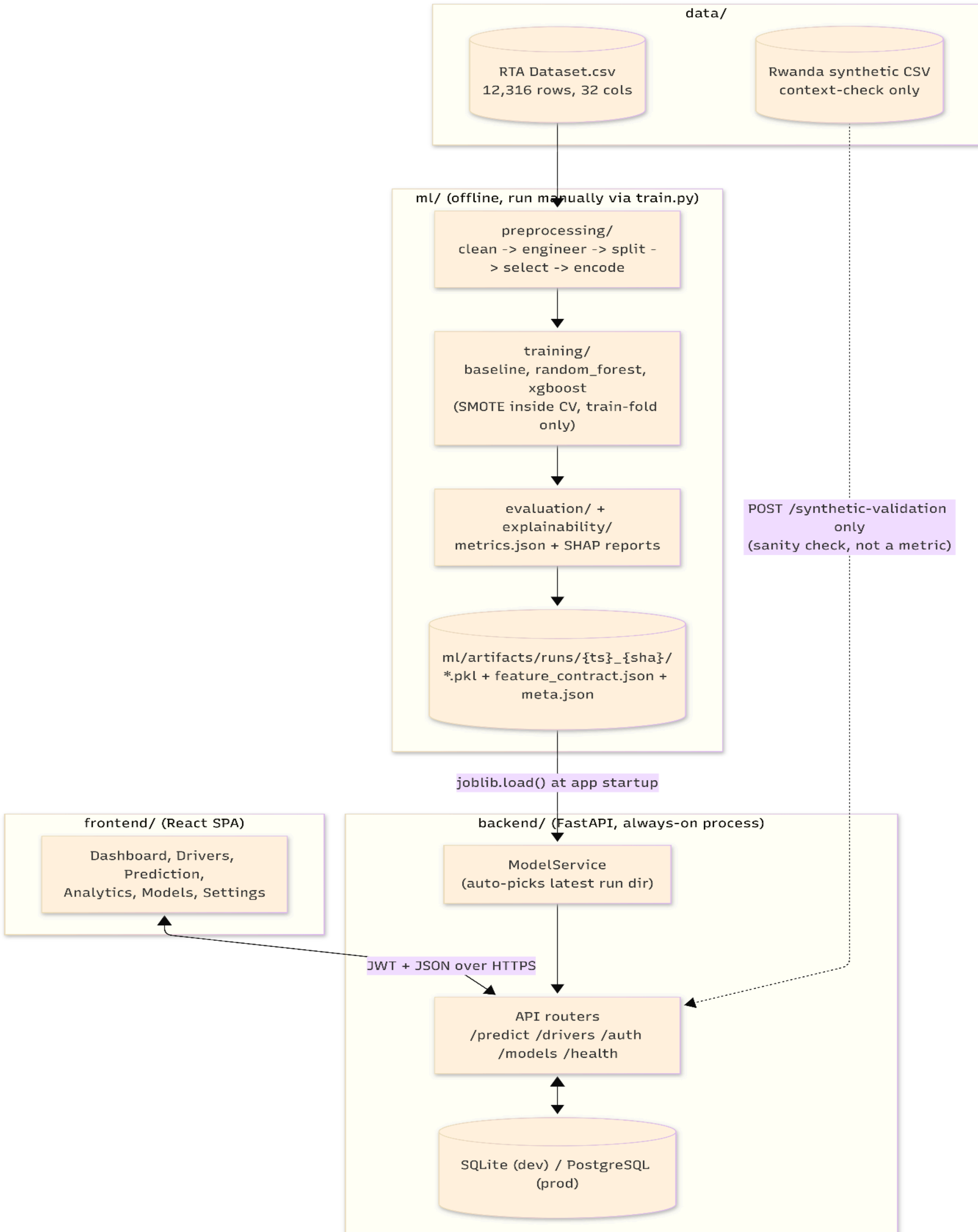


Figure 2: Data pre-processing, Proposed MBERE ML methodology and machine-learning pipeline.

3.3.1 Baseline, models and evaluation metrics

The rule-based baseline is a hand-authored additive risk score mapped to class probabilities; it is explicitly not a most-frequent dummy classifier, and the machine-learning models are required to beat it on the headline metrics. Because of the extreme imbalance, accuracy is deprioritized (a model can reach ~85% accuracy by predicting Slight for everyone). The headline metrics are therefore macro-F1, macro-recall and one-vs-rest macro ROC-AUC, reported alongside per-class precision, recall and F1 with particular attention to Fatal-class recall a confusion matrix, and ROC curves. It is stated frankly that correctly recalling the rare Fatal class from profile-level features is very hard and is a known limitation to be reported rather than concealed.

3.3.2 Definition of the driver risk score and its calibration status

The driver risk score is a probability-weighted expected-severity index, not a calibrated crash probability. For the three-class case it is computed as the dot product of the class index and the predicted class probabilities, normalised to the unit interval; equivalently,

Risk = (P(Serious) + 2·P(Fatal)) / 2, which lies in [0, 1].

The score is then banded into Low, Medium and High using two fixed thresholds (0.34 and 0.67). Two honest caveats follow. First, the banding thresholds are a transparent business rule, not learned from data. Second, no probability calibration (for example Platt scaling or isotonic regression) is applied, and because SMOTE can degrade calibration (van den Goorbergh et al., 2022) the score should be interpreted as a relative ranking signal for triage and monitoring rather than as a statistically calibrated probability. Adding and evaluating calibration is identified as a priority enhancement before any pricing-adjacent use.

3.3.3 Retraining Loop

The methodology diagram (figure 2) shows a retraining loop that would return new labelled outcomes to the training stage. This is prototype/future design, not an implemented capstone feature: it is only feasible with a confirmed data-sharing partner supplying new labelled outcomes. It is documented here as an architectural intention and clearly marked as future work.

3.3.4 Use Case Diagram

Figure 3 models the actors that interact with the platform and the functions available to each. Three actors are distinguished. The Insurer uses the system as a risk-based-pricing input and for portfolio risk monitoring; the Fleet Manager uses it to target driver safety interventions and training check-ins; and the System Administrator handles model and platform operations. Both operator roles authenticate, manage driver records, request driver risk assessments (each of which includes a SHAP explanation returned in the same call), view a driver's risk history, and view fleet or portfolio analytics. The System Administrator additionally manages user accounts and access, trains and registers new model versions, configures and deploys the served model, monitors system health and the model registry, and can run the synthetic-context validation check. The «include» relationship shown from "Request driver risk assessment" to "View SHAP explanation" reflects that every prediction is accompanied by a per-prediction explanation, in keeping with the interpretability objective.

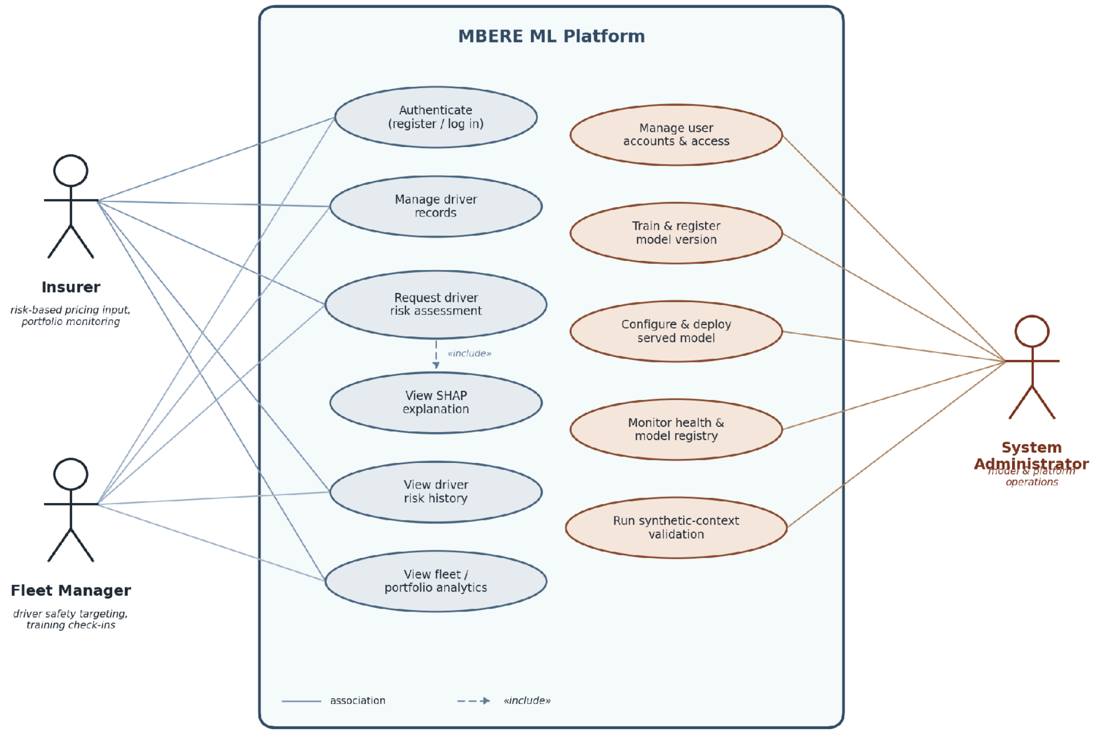


Figure 3: System use case diagram (Insurer, Fleet Manager, System Administrator).

3.5 Class Diagram

Figure 4 shows the principal classes of the platform and their relationships. The Driver class holds driver attributes; FeatureRecord captures the engineered driver-and-context features used for prediction; RiskModel encapsulates training, prediction and explanation; and RiskAssessment records a computed severity-risk score, band and the model version used. The User and Dashboard classes represent the insurer or fleet operator and the interface through which they view and analyse assessments.

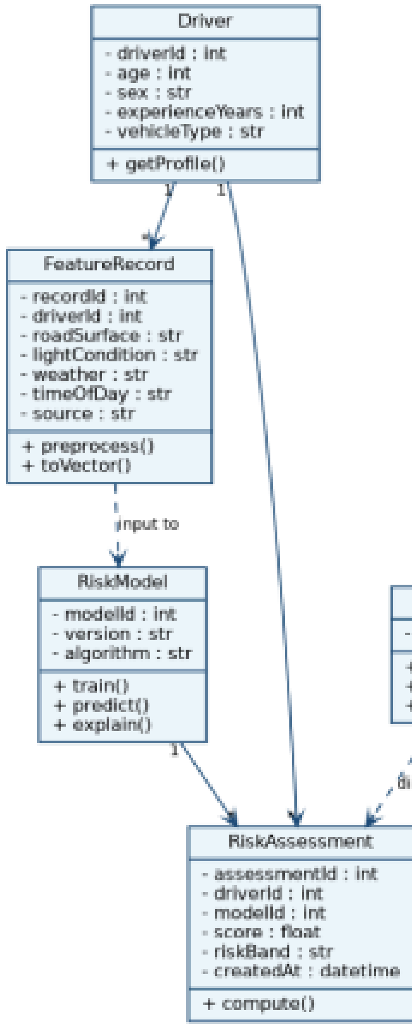


Figure 4: System class diagram.

This class model describes the prototype's persistence layer and its intended future integration with insurer/fleet systems. The Addis Ababa training data are cross-sectional crash records without stable driver identities or longitudinal histories, so persistent per-driver profiles are a property of the platform (populated by operator-entered or synthetic records), not of the training dataset. The Driver and RiskAssessment entities should therefore be read as prototype/future-integration constructs rather than as entities recovered from the source data.

3.6 System Architecture

The architecture, shown in Figure 5, is organized into four layers. The data layer stores the crash-record dataset and the synthetic Rwandan-context sample, together with a feature store of engineered features. The machine-learning layer holds the trained model (Random Forest or XGBoost), an inference API that returns severity-risk scores, and a SHAP explainer. The application layer is a web dashboard presenting a driver list, individual risk scores, per-driver explanations and a portfolio view. The users are motor insurers and fleet or motorcycle-taxi operators. A retraining pipeline is shown feeding new labelled outcomes back into the feature store and model; as noted in **Section 3.3.3**, this is a future-work component.

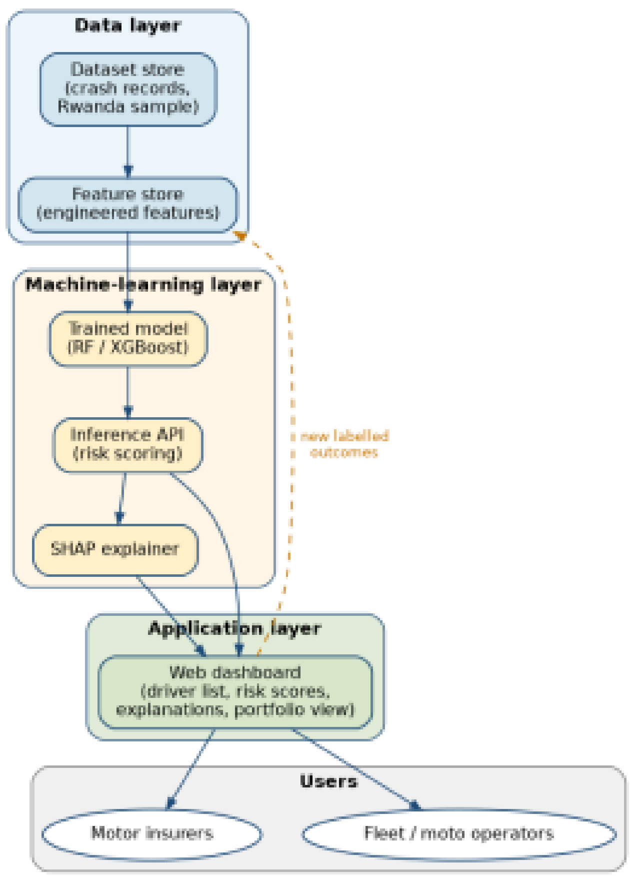


Figure 5: MBERE ML system architecture.

Data-sharing and guardrails. Acquisition of any Rwandan institutional sample is not guaranteed; it would require formal approval and a data-sharing agreement with RNP/RBC, and if it is not obtained the system proceeds on Addis data with synthetic-context checking only

(Section 3.2.4). The architecture embeds explicit guardrails: no automatic premium setting; every prediction carries an explanation and a confidence-limited interpretation; and a fairness audit is run across age, sex, vehicle type and the motorcycle category before any operational recommendation is surfaced.

3.7 Entity-Relationship Diagram

The entity-relationship diagram in Figure 6 models the data the platform persists. A DRIVER has many FEATURE_RECORD entries and receives many RISK_ASSESSMENT records over time; each MODEL_VERSION produces many risk assessments, allowing predictions to be traced to the model that generated them; and each APP_USER (an insurer or fleet operator) reviews many risk assessments. This structure supports auditability and versioning.

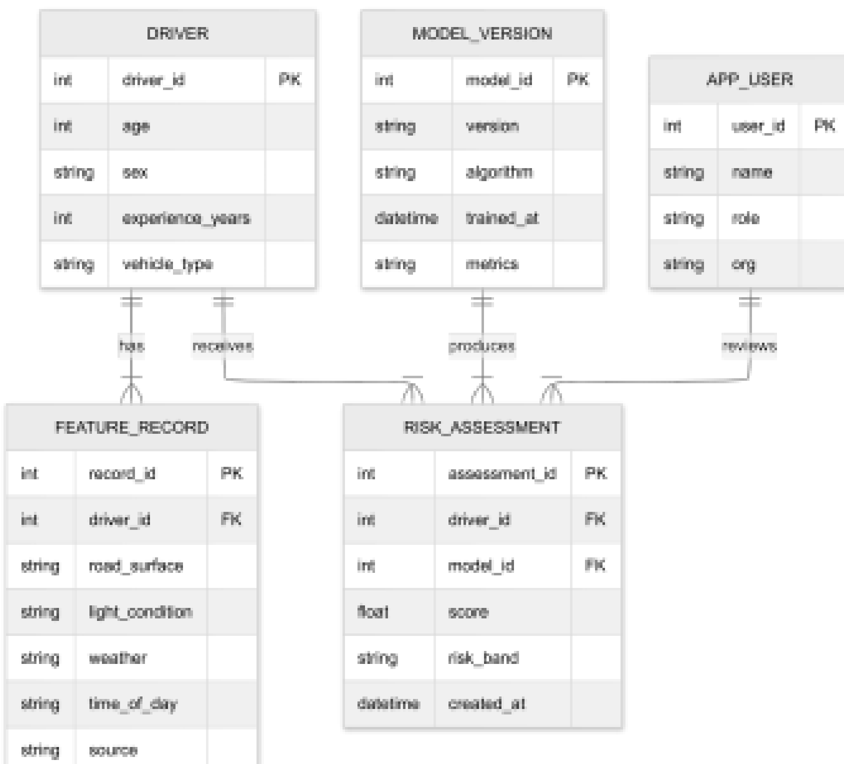


Figure 6: Entity-relationship diagram (ERD).

This ERD supports a longitudinal decision-support platform, whereas the available training dataset is not a longitudinal driver history. The schema should therefore be understood as the prototype's persistence design and a target for future insurer/fleet integration; for the capstone it is populated by operator-entered records and per-prediction audit rows rather than by recovered driver timelines.

3.8 Ethics and-Responsible Use

Because severity-risk scores can affect premiums, employment and access to work, responsible-use safeguards are treated as part of the design, not an afterthought.

- **Data protection and anonymization.** The Addis Ababa dataset contains no personal identifiers, and the synthetic-context sample is artificial by construction. Any future Rwandan institutional data would require permissions, a data-protection review and possibly ethics approval, with anonymization and appropriate consent or waiver in place before use.
- **Access control.** The platform requires authentication, and administrative functions (model training, deployment, user management) are separated from operator functions, as reflected in the use case model.
- **Human-in-the-loop and disclaimers.** Outputs are decision support only. Every score is presented with its SHAP explanation and a clear disclaimer that it is a relative severity-risk ranking—not a calibrated probability and not a pricing decision—and requires human review.
- **Fairness and proxy-discrimination audit.** Following Lindholm et al. (2022) and Prince & Schwarcz (2019), removing protected attributes is insufficient because other features can act as proxies. The project therefore prunes weak/sensitive inputs (for example driver sex), and audits outcomes across age, sex, vehicle type and the motorcycle category to check for disparate impact.
- **Recommendation on driver attributes in pricing.** The project explicitly recommends that driver attributes not be used for automatic insurance-pricing decisions on the basis of this model; any pricing use would require actuarial and claims validation, calibration, and regulatory review beyond the capstone's scope.

3.9 Development Tools

- **Data and machine learning:** Python with pandas and NumPy for data preparation, scikit-learn for the rule-based baseline and Random Forest, the XGBoost library for gradient boosting, imbalanced-learn for SMOTE (applied inside training folds), and SHAP for explainability.
- **Datasets:** the Addis Ababa Road Traffic Accident dataset (Bedane, 2020) as the sole source of reported performance, a Tonic's Fabricate synthetic Rwandan-context sample for functionality and robustness checking only, and a requested but non-blocking Rwandan validation sample from RNP and RBC.
- **Back-end and data storage:** a REST API built with FastAPI and a relational database (SQLite for development, PostgreSQL for production) for drivers, feature records, model versions and risk assessments, with each prediction persisted as an auditable trail against the model version that produced it.
- **Web platform:** a lightweight React/TypeScript front-end presenting the driver list, risk scores, SHAP-based explanations and portfolio view; the front-end reads the model's feature contract so it need not change when the served feature set changes.
- **Environment and tooling:** Git and GitHub for version control, versioned run directories for experiment tracking, and Matplotlib and Graphviz for analysis figures and diagrams.

The machine-learning contribution target definition, baseline and model comparison, imbalance handling, calibration awareness, explainability and limitations remains the centre of the work; the front-end is a delivery vehicle for that contribution rather than the focus. Any use of institutional data would be subject to the data-protection and ethics safeguards of **Section 3.8**.

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